

Document Name: Master Batch Record: Downstream Purification (Affinity Capture and Anion Exchange)			Page 1	
Number: MBR-DSP-201	Version: 1.0	Effective Date: 01APR2026	Status: Effective	Part No: DS-OBX-201

Record Reproduction Certification

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Product and Lot Information

Product	OBX-201 (recombinant AAV9)	Drug substance lot	DS-26052
Process step	Affinity capture and anion exchange polish	Manufacturing date	20JUN2026
Affinity resin	POROS CaptureSelect AAVX	Affinity column ID	AFF-COL-02
AEX medium	Strong anion exchange monolith (Q)	AEX column ID	AEX-COL-01
Clarified harvest lot	HAR-26052	Target percent full	NLT 80.0 by AEX-HPLC

Referenced SOPs

SOP Number	Title
SOP-3300	Affinity Capture Chromatography (CaptureSelect AAVX)
SOP-3320	Low-pH Elution and Neutralization
SOP-3400	Anion Exchange Full and Empty Separation
SOP-3410	Fraction Collection and Pooling
SOP-1203	Buffer and Solution Preparation
SOP-1318	In-Process UV, Conductivity, and pH Monitoring
SOP-0905	Bioburden and Endotoxin Sampling

1. EQUIPMENT / INSTRUMENTS			
Description	EIN	Calibration Due	Performed By / Date
Chromatography skid	SKID-DSP-01	30SEP2026	SK 20JUN26
Inline UV detector	UV-231	12OCT2026	SK 20JUN26
Inline conductivity monitor	CON-128	05NOV2026	SK 20JUN26
Inline pH meter	PHM-214	20AUG2026	SK 20JUN26
Fraction collector	FRC-04	18OCT2026	SK 20JUN26
Analytical balance	BAL-021	22NOV2026	SK 20JUN26

2. BUFFER PREPARATION

Buffer	Composition	Lot Number	Prepared By / Date	Verified By / Date
Affinity equilibration	20 mM sodium phosphate, 2 mM MgCl ₂ , pH 5.0	BUF-AEQ-0605	SK 20JUN26	PA 20JUN26
Affinity elution	100 mM sodium citrate, 2 mM MgCl ₂ , pH 3.0	BUF-AEL-0605	SK 20JUN26	PA 20JUN26
Neutralization buffer	1 M Tris base, pH 8.5	BUF-NEU-0605	SK 20JUN26	PA 20JUN26
AEX equilibration	20 mM Bis-Tris-propane, pH 9.0	BUF-QEQ-0605	SK 20JUN26	PA 20JUN26
AEX elution (high salt)	20 mM Bis-Tris-propane, 1 M TMAC, pH 9.0	BUF-QEL-0605	SK 20JUN26	PA 20JUN26

3. AFFINITY CAPTURE CHROMATOGRAPHY

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
3.1	Record clarified harvest load volume.	Load volume (L). Acceptance: record.	9.7	SK 20JUN26	PA 20JUN26
3.2	Record equilibration conductivity.	Conductivity (mS/cm). Acceptance: 8 to 12.	10.1	SK 20JUN26	PA 20JUN26
3.3	Confirm baseline stable before load.	Baseline stable?	☒ X Yes ☐ No	SK 20JUN26	PA 20JUN26
3.4	Record affinity elution pH.	Elution pH. Acceptance: 2.8 to 3.2.	3.0	SK 20JUN26	PA 20JUN26
3.5	Record low-pH hold time before neutralization.	Hold time (min). Acceptance: not more than 15.	27	SK 20JUN26	PA 20JUN26
3.6	Record neutralized pool pH.	pH. Acceptance: 7.0 to 8.0.	7.5	SK 20JUN26	PA 20JUN26
3.7	Record affinity eluate pool volume.	Volume (L). Acceptance: record.	1.9	SK 20JUN26	PA 20JUN26

4. ANION EXCHANGE FULL AND EMPTY SEPARATION

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Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
4.1	Record anion exchange load conductivity.	Conductivity (mS/cm). Acceptance: not more than 5.0 to bind.	6.8	SK 20JUN26	PA 20JUN26
4.2	Record anion exchange load pH.	pH. Acceptance: 8.8 to 9.2.	9.0	SK 20JUN26	PA 20JUN26
4.3	Confirm gradient method loaded.	Method AEX-QA-01. Empty peak about 6.3 mS/cm; full peak about 10.1 mS/cm.	☒ X Yes ☐ No	SK 20JUN26	PA 20JUN26
4.4	Record fraction-collection (pooling) start conductivity.	Conductivity (mS/cm). Acceptance: not less than 8.5 to collect the full-capsid peak and exclude the empty shoulder.	7.2	SK 20JUN26	PA 20JUN26
4.5	Record fraction-collection stop conductivity.	Conductivity (mS/cm). Acceptance: record.	13.1	SK 20JUN26	PA 20JUN26
4.6	Record anion exchange pool volume.	Volume (L). Acceptance: record.	1.7	SK 20JUN26	PA 20JUN26
4.7	Record anion exchange load total vector genomes.	Load VG (e13 vg). Acceptance: record.	5.00	SK 20JUN26	PA 20JUN26
4.8	Record anion exchange pool total vector genomes.	Pool VG (e13 vg). Acceptance: record.	3.50	SK 20JUN26	PA 20JUN26

5. POOL AND STEP RECOVERY

Step	Instruction	Parameter	Actual	Performed By / Date	Verified By / Date
5.1	Calculate anion exchange step recovery.	Step recovery (percent). Acceptance: not less than 55 percent. Formula: Step recovery (percent) = (Pool VG / Load	Pool VG (e13 vg): 3.50 Load VG (e13 vg): 5.00 Result (percent): 70.0	SK 20JUN26	PA 20JUN26

		VG) x 100			
5.2	Record percent full capsids confirmed on the analytical worksheet.	Percent full (percent), reference; confirmed on WS-AEX-26052. Acceptance: not less than 80.0.	72.4	SK 20JUN26	PA 20JUN26
5.3	Confirm purified pool bioburden sample collected.	Bioburden sample collected?	☒ X Yes ☐ No	SK 20JUN26	PA 20JUN26

6. PROCESS COMMENTS			
Page / Step	Comment	Performed By / Date	Verified By / Date
Step 3.5	Low-pH hold extended to 27 min against the not-more-than-15-min limit after a delay staging neutralization buffer. Capsid integrity risk noted for review. Ref DEV-26-0209.	SK 20JUN26	PA 20JUN26
Step 4.1	Anion exchange load conductivity 6.8 mS/cm, above the not-more-than-5.0 mS/cm target; the harvest dilution step under-diluted the load. Ref DEV-26-0210.	SK 20JUN26	PA 20JUN26
Step 4.4	Fraction collection started at 7.2 mS/cm, below the not-less-than-8.5 mS/cm pooling window, drawing part of the empty-capsid shoulder into the pool. Percent full on WS-AEX-26052 subsequently 72.4 percent, below the 80.0 percent specification. Ref DEV-26-0207.	SK 20JUN26	PA 20JUN26

BATCH RECORD APPROVALS	
Manufacturing Review	Quality Assurance Review
F. Castellano 16JUL2026	L. Marchetti 16JUL2026

Manufacturing and QA review signatures are applied at batch disposition.